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Name.....

Reg. No.....

**THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2025**

Economics

ECO 4 (3) C04—MATHEMATICAL TOOLS FOR ECONOMICS—II

(2020—2023 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A (Short Answer Questions)*Maximum marks in this Section is 25.**Students can attempt **all** questions.**Each question carries a maximum of 2 marks.*

1. Continuous function.
2. Prove the addition rule of derivative, $y = x^3 + 6x^2 - 5x + 20$.
3. Relative extrema.
4. Marginal utility.
5. Slope of a curvilinear function.
6. Implicit and explicit differentiation.
7. Marginal productivity.
8. Optimize the function $x = 2x^3 - 30x^2 + 126x + 59$.
9. Euler's theorem.
10. Limit of a function.
11. Integration by substitution.
12. Constrained and unconstrained optimization.
13. Optimal solution.
14. Short run and long run production function.
15. Definite integral.

Turn over

Section B (Short Essay/Paragraph Questions)

Maximum marks in this Section is 35.

*Students can attempt **all** questions.*

Each question carries a maximum of 5 marks.

16. Distinguish between increasing and decreasing function. Check whether the function $3x^3 + 3x^2 + x - 10$ is increasing or decreasing.
17. Explain various marginal concepts in economics.
18. Explain linear homogeneous production function. Write a note on Cobb Douglas production function.
19. What do you mean by derivative of a function ? Find the derivative of the following function using the definition of the derivative $f(x) = 4x^2 + 100x + 20$.
20. Differentiate between AC and MC. From the total cost function $TC = Q^3 - 5Q^2 + 60Q$ find :
 - i) The AC function ;
 - ii) The critical value at which AC is minimized ; and
 - iii) The minimum average cost.
21. Describe the procedure to find the area between a curve and a line using integration.
22. Define curve sketching. Describe general guidelines for curve sketching.
23. *Lagrange multiplier is a strategy for finding the local maxima and minima of a function subject to equality constraints - Examine.*

Section C (Long Essay Questions)

*Answer any **two** questions.*

Each question carries a maximum of 10 marks.

24. Explain power rule, sum rule and difference rule of integration. Integrate the following as per the above rules respectively
 - i) $\int x^3 dx$.
 - ii) $\int (x + x^2) dx$.
 - iii) $\int (x - x^2) dx$.

25. Explain the different kinds of elasticity and their estimation process using numerical examples
26. Narrate the optimization of multivariable functions in economics. Assume that a firm producing two goods x and y has the profit function.

$$\Pi = 64x - 2x^2 + 4xy - 4y^2 + 23y - 14.$$

Find the profit-maximizing level of output for each of the two goods.

27. Examine the uses of derivatives in economics. Describe meaning and measurement of concavity and convexity of a function.

(2 × 10 = 20 marks)